

Machine Vision Module for Stage Recognition and Active Brazing Event Detection in Induction Brazing of Spacecraft Waveguide Paths

A. V. Ilin

*Department of Applied Artificial Intelligence
RUDN University
Moscow, Russian Federation
a.ilin.v@yandex.ru*

E.V. Shutkina

*Research and Education Center
Bauman Moscow State Technical University
Moscow, Russian Federation
elizaveta-shutkina@mail.ru*

Abstract — Induction brazing of spacecraft waveguide paths requires tight control of technological stages and accurate timing of the start of active brazing. Existing thermocouple- and pyrometer-based control loops describe the heating regime well but observe the local visual state of flux, filler metal and seam only indirectly.

This paper presents a machine-vision module that classifies four visually distinguishable stages — preparation heating, flux activation, active brazing, and stabilization. In addition, the module detects the onset of active brazing and generates a temperature-hold recommendation. The module combines region-of-interest extraction, a convolutional classifier, a technologically constrained state machine, and an event trigger built on the probability of the active brazing stage.

An interpretable OpenCV baseline and a ResNet18 classifier are trained on 3 FPS and 10 FPS frame datasets split by video identifier; MobileNetV3 Small and ResNet18 with 64×64 input are evaluated as CPU-friendly alternatives. ResNet18 reaches macro-F1 0.8209 on the full 10 FPS test set and 0.9175 when excluding one visually ambiguous hard-case recording. The neural trigger detects the active-brazing onset on all test videos with a mean absolute time error of 0.70 s. The 50 ms-per-frame latency target on CPU is met by MobileNetV3 Small and by ResNet18 with 64×64 input.

Keywords — *induction brazing; spacecraft waveguide paths; machine vision; stage classification; state machine; event detection; real-time inference.*

I. INTRODUCTION

Spacecraft communication and radio-technical systems depend on waveguide paths whose electromagnetic characteristics must remain stable over the device's operational life. During assembly, joint quality is determined by the temperature regime, flux state, filler-metal melting, flow into the seam, and subsequent hold at temperature [1], [2]. Induction brazing concentrates heat in the joint zone, but the process has a narrow window: underheating leaves seam defects, while overheating distorts the waveguide geometry [3], [4]. Conventional control relies on thermocouple or pyrometer

measurements, an inductor power program, and operator experience [5], [6]. These signals describe the heating regime, but they only indirectly reflect the local visual state of flux, filler metal, and seam, where the start of active brazing is actually observed.

Deep convolutional networks have become standard tools for industrial visual inspection [7]–[10]. Residual architectures [11], modern frameworks [12] and classical OpenCV features [13] together cover the learning and interpretable sides of the task, while MobileNetV3 [14] and the wider edge-inference trend [15] indicate that such models can run on CPU within tight latency budgets. Earlier work on the control of induction brazing of waveguide paths focused on indirect temperature measurements and on automated control loops [16], and on neural-network process models [17]. Comparatively little attention has been paid to a self-contained visual channel that observes the brazing zone, classifies its stage and produces an event signal aligned with the start of active brazing.

This setting also determines the choice of models. Hand-crafted OpenCV features are kept as an interpretable baseline: they make it possible to check whether brightness, highlights, texture and frame-to-frame changes are sufficient for separating the stages. ResNet18 is used as the main quality-oriented convolutional classifier, since the task requires learned spatial features over the ROI rather than only manually specified image statistics. MobileNetV3 Small and a reduced-input ResNet18 variant are evaluated as CPU-friendly alternatives, because the final module must operate close to real time and meet the 50 ms-per-frame latency target on limited hardware.

This work addresses this gap with a machine-vision module that extracts an ROI from the brazing-zone video stream, classifies each frame, stabilises the operator-facing stage through a state machine, and generates active_brazing_started for the HOLD TEMPERATURE recommendation.

II. PROBLEM STATEMENT

The brazing process is decomposed into four visually distinguishable stages in fixed technological order: «inactive_preparation» (set-up and pre-heating),

«flux_activation» (the flux reaches working temperature, becomes transparent and prepares the surface for wetting by the filler metal), «active_brazing» (the filler metal melts and flows into the seam, defining «active_brazing_started»), and «stabilization» (the joint is held in the achieved regime). Two computational tasks are defined on top of these stages.

Stage labels are assigned from interval annotations with video identifier, stage name, and start/end timestamps, then converted into frame-level labels after extraction.

The first task is frame-level classification: for each frame the module outputs a probability distribution over the four stages. The classes are imbalanced — flux_activation and active_brazing contain noticeably fewer frames than inactive_preparation and stabilization — and the short stages are the most technologically informative, so the primary metric is macro-F1, supported by weighted-F1, accuracy and per-class F1 and recall on active_brazing.

The second task is event detection. From the frame-level probability of active_brazing the module generates a binary signal active_brazing_started whose timing is close to the human-annotated start of the stage. For a video with annotated start t_{true} and trigger response t_{pred} the event error is $e = t_{\text{pred}} - t_{\text{true}}$; positive values indicate late detection, negative early detection. The aggregate trigger metrics are the shares of detected, early and late activations, the mean absolute error, and the shares of videos with absolute error within 1 and 2 seconds.

The input is a region of interest cropped from a fixed rectangle around the seam zone; the rest of the frame is excluded. Two datasets are used: 3 FPS for the exploratory experiments and 10 FPS for the final classifier and the state machine. Train, validation and test partitions are formed by video identifier to prevent leakage of adjacent frames between subsets. Recording MVI_6265 (Fig. 1) is treated as a hard case (pronounced highlights, ambiguous flux behaviour, atypical filler metal dynamics) and kept in the test set as a stress test.



Fig. 1. Hard-case example from MVI_6265 with pronounced highlights and ambiguous flux/filler-metal dynamics.

For the final 10 FPS dataset, the split contains 5,755 training frames, 1,091 validation frames and 1,732 test frames. The final dataset contains 8,578 frames from 15 recordings after excluding three blacklisted recordings with atypical visual conditions. The class distribution is shown in Table I: inactive_preparation and stabilization account for 78.5% of the

frames, while flux_activation and active_brazing together account for 21.5%.

TABLE I. CLASS DISTRIBUTION IN THE FINAL 10 FPS DATASET

Class	Train	Validation	Test	Total
inactive_preparation	3,552	734	1,000	5,286
flux_activation	506	102	138	746
active_brazing	766	138	196	1,100
stabilization	931	117	398	1,446
Total	5,755	1,091	1,732	8,578

This distribution explains why aggregate accuracy alone is insufficient: the rare transition stages carry most of the value for event detection.

Three practical requirements follow. First, aggregate accuracy is insufficient if it hides poor active_brazing recall, because this class drives the event signal. Second, the operator-facing stage must obey the technological order and must not return from active_brazing to flux_activation after a single erroneous frame. Third, the full pipeline, including frame read, ROI extraction, preprocessing, network inference, trigger, and post-processing, must stay within a 50 ms-per-frame target.

III. PROPOSED METHOD AND SYSTEM ARCHITECTURE

A. Pipeline overview

The module is organised as a linear pipeline: a frame is read; the fixed ROI rectangle is cropped (Fig. 2), resized and normalised; the classifier outputs a probability vector over four stages; temporal post-processing yields the operator-facing Stable Stage; in parallel, P(active_brazing) feeds the event trigger that may generate active_brazing_started and the demonstration HOLD TEMPERATURE recommendation. The same module serves both the offline evaluation mode and the demonstration mode that overlays Raw Stage, Stable Stage, P(active_brazing), trigger states, inference time and the ROI rectangle on the source video.

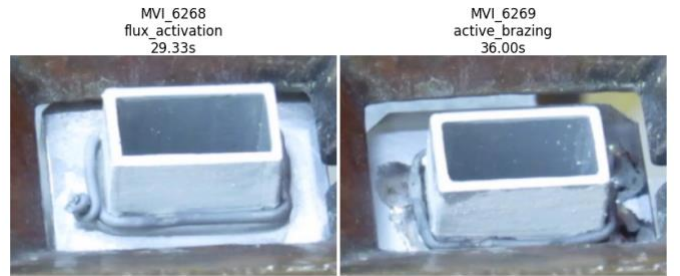


Fig. 2. Examples of cropped regions of interest used by the classifier: flux_activation in MVI_6268 and active_brazing in MVI_6269.

B. Interpretable OpenCV baseline

An interpretable baseline uses OpenCV-style hand-crafted features: brightness statistics, dominant colour components, texture energy, density of bright highlights, edge density, and frame-to-frame differences. The feature vectors train logistic regression, random forest and gradient boosting models; the

strongest variant is logistic regression with majority temporal smoothing [18]. The baseline plays two roles: an interpretable reference point for deep models, and a source of the most informative features that are reused later as inputs of a fallback CV-based event trigger.

C. Neural stage classifier

The main classifier is a convolutional network on the ROI patch. ResNet18 is used as the quality-oriented model, while MobileNetV3 Small and ResNet18 with 64×64 input are evaluated as CPU-friendly alternatives under the latency budget. Class imbalance is handled with weighted cross-entropy; training uses AdamW with decoupled weight decay [19], moderate image augmentations [20], PyTorch, and validation-based checkpoint selection by macro-F1. All neural experiments use ROI inputs and batch-size-1 inference. The qualitative ResNet18 model uses 224×224 inputs, while the CPU-oriented ResNet18 variant uses 64×64 inputs. F1 and recall on active_brazing are tracked as secondary criteria because this class drives the event trigger.

D. Stabilisation through a state machine

Raw per-frame predictions are accurate on aggregate but unstable near stage boundaries. Three post-processing variants are compared: raw predictions, majority smoothing, and a state machine. Majority smoothing replaces each prediction by the most frequent class within a sliding window, but it does not enforce technological order. The state machine follows the fixed chain inactive_preparation → flux_activation → active_brazing → stabilization. A transition is accepted only if at least min_confirm_frames recent predictions support the next stage within a history of length window_size. The selected configuration uses min_confirm_frames = 5 and window_size = 7; backward transitions are forbidden by construction.

E. Event trigger for active brazing

The event signal active_brazing_started is built on P(active_brazing) from the classifier. The trigger fires when the probability exceeds a threshold on a configurable number of consecutive frames. Two operating points are used: a research configuration with threshold 0.2 and seven-frame confirmation aimed at robust event timing with few early false alarms, and a demonstration configuration with threshold 0.4 and single-frame confirmation that minimises reaction delay. Two backup variants are also analysed: an OpenCV rule-based trigger on the same interpretable score as the baseline, and a binary classical ML trigger on OpenCV features. The neural trigger is the main signal; the CV rule trigger remains as an interpretable fallback. The same classifier produces both the per-frame stage and the event-driving probability, keeping the architecture single-model.

IV. EXPERIMENTAL EVALUATION

A. Protocol and data

All variants follow the protocol defined in Section II. The OpenCV baseline and the neural baseline are evaluated on 3 FPS; the final classifier, state machine, and event trigger are evaluated on the 10 FPS test set. Selected results are reported

with and without hard-case MVI_6265. Latency is measured over the full processing cycle, not only the network forward pass.

B. Stage classification results

Table II summarises per-frame results for the baseline, the 3 FPS neural model, the final ResNet18 on 10 FPS with and without temporal smoothing, and the same model without the hard-case recording.

TABLE II. PER-FRAME CLASSIFICATION METRICS ON THE TEST SET

Model / mode	Accuracy	Macro-F1	Weighted-F1
OpenCV baseline (LogReg + smoothing)	0.6296	0.5809	0.6964
ResNet18, 3 FPS	0.9347	0.8489	0.9307
ResNet18 base, 10 FPS — raw	0.8747	0.8209	0.8895
ResNet18 base, 10 FPS — temporal smoothing	0.8776	0.8290	0.8926
ResNet18 base, 10 FPS — raw, no MVI_6265	0.9598	0.9175	0.9573
ResNet18 base, 10 FPS — smoothing, no MVI_6265	0.9624	0.9234	0.9601

The OpenCV baseline saturates around macro-F1 0.58 even after smoothing: hand-crafted features register that the ROI is changing but do not separate flux_activation from early active_brazing, the two stages that share rising brightness, highlights and texture drift. The 3 FPS neural baseline already lifts macro-F1 to 0.85, motivating the move to 10 FPS for finer temporal resolution of short stages. The final ResNet18 on 10 FPS reaches macro-F1 0.8209 on the full test set; with temporal smoothing macro-F1 rises to 0.8290 and active_brazing F1 to 0.8368 with recall 0.7194. Excluding MVI_6265 lifts macro-F1 to 0.9175 (0.9234 with smoothing) — the residual gap on the full set is dominated by one difficult scenario, not a methodological limitation.

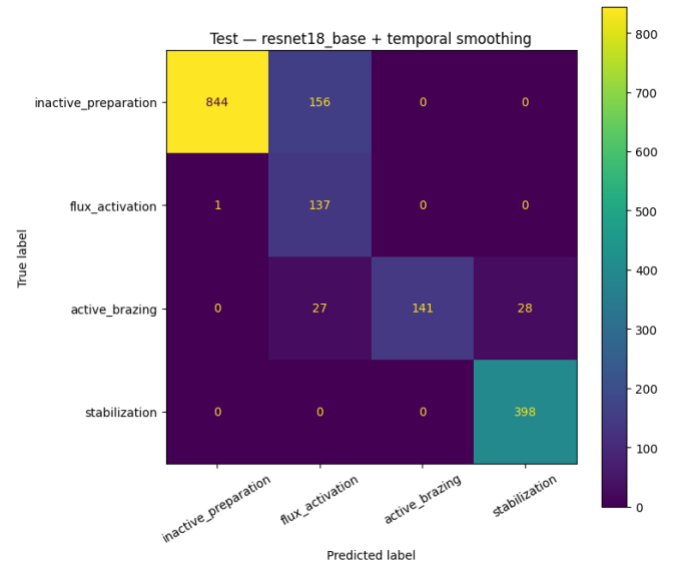


Fig. 3. Confusion matrix of ResNet18 base with temporal smoothing on the 10 FPS test set.

Fig. 3 confirms that errors are concentrated at stage boundaries. The most frequent mistakes are inactive_preparation $\rightarrow$ flux_activation in the early phase, active_brazing $\rightarrow$ flux_activation and active_brazing $\rightarrow$ stabilization around the start and end of melting, and occasional stabilization $\rightarrow$ active_brazing later. Stable states are recognised reliably; the difficulty lies in transitions where visual features overlap continuously rather than switch discretely. A small ablation showed that the active_brazing-boosted ResNet18 variant raised the recall of that class but lost ground on macro-F1 (validation macro-F1 0.9005 against 0.9199 for the base model), confirming that an aggregate-balanced loss is preferable when the trigger consumes the class probability directly.

C. Stage stabilisation

Table III compares the three post-processing variants on the full test set and on the subset without MVI_6265. Majority smoothing slightly improves per-frame metrics over raw predictions, as expected for a method optimising local agreement with labels. State machine trades a small amount of per-frame quality for technological consistency: it forbids backward transitions and produces a stage sequence matching the physics of the process — the property needed for an operator display.

TABLE III. POST-PROCESSING VARIANTS FOR THE RESNET18 STAGE CLASSIFIER

Method	Accuracy	Macro-F1	Weighted-F1	Active brazing F1	Active brazing recall
Raw predictions, full test	0.8747	0.8209	0.8895	0.8097	0.6837
Majority smoothing, $w = 5$, full	0.8776	0.8290	0.8926	0.8368	0.7194
State machine, full	0.8441	0.7988	0.8674	0.8304	0.7245
Raw, no MVI_6265	0.9598	0.9175	0.9573	0.8051	0.6786
Smoothing, no MVI_6265	0.9624	0.9234	0.9601	0.8186	0.6929
State machine, no MVI_6265	0.9581	0.9155	0.9562	0.8333	0.7143

Without MVI_6265 state machine reaches accuracy 0.9581, macro-F1 0.9155 and weighted-F1 0.9562 — on a par with raw predictions and smoothing while guaranteeing a monotone operator stage. A grid search over min_confirm_frames $\in \{3, 5, 7\}$ and window_size $\in \{7, 10, 15\}$ showed that smaller min_confirm_frames yields marginally higher per-frame metrics (e.g. accuracy 0.8493 and macro-F1 0.8083 at min_confirm_frames = 3 across all window sizes), but the more conservative configuration was retained to suppress premature transitions on isolated prediction spikes. Fig. 4 shows the effect: short local errors of raw predictions are filtered out and the operator stage follows a clean monotone trajectory.

D. Active brazing event trigger

Event detection is evaluated separately. For each test recording the annotated start of active_brazing is compared

with the first sustained activation of each trigger; the three variants appear in Table IV.

TABLE IV. EVENT-LEVEL COMPARISON OF ACTIVE BRAZING TRIGGER VARIANTS ON THE TEST RECORDINGS.

Trigger	Detected	Early	Late	MAE, s	Max error, s	within 1 s	within 2 s
P(active_brazing), thr 0.2, 7 frames	3 / 3	0	3	0.70	1.0	1.00	1.00
OpenCV rule, thr 0.5, 5 frames	3 / 3	1	2	0.90	1.8	0.67	1.00
CV binary ML, thr 0.6, 7 frames	2 / 3	0	2	0.95	1.5	0.50	1.00

Three trigger variants are compared. The neural trigger uses P(active_brazing) from the ResNet18 classifier and fires after sustained evidence above the selected threshold. The OpenCV rule trigger uses an interpretable hand-crafted score based on visual changes in the ROI. The binary classical ML trigger uses OpenCV-derived features but learns the event decision as a separate binary classifier.

The neural trigger on P(active_brazing) detects the event on all three recordings with mean absolute error 0.70 s, maximum 1.0 s, and 100% of detections within both 1 s and 2 s tolerance. The OpenCV rule trigger also fires on all recordings but produces one early detection and a larger maximum error of 1.8 s; the binary classical ML trigger misses one of three. Sensor-based ensemble evidence in the literature supports the choice of a probability-driven trigger over a hand-crafted score.

Because the event-level evaluation uses three annotated test recordings, the trigger metrics should be interpreted as a controlled prototype check of the proposed event logic. They show how the logic behaves on the available test videos, but they should not be treated as a final statistical estimate for industrial deployment.

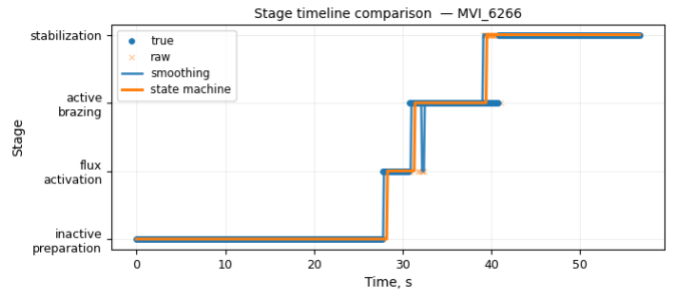


Fig. 4. Stage timeline on the test recording MVI_6266: ground truth, raw ResNet18 predictions, majority smoothing and state machine

The neural trigger is therefore used as the main source of active_brazing_started, while the OpenCV rule trigger remains an explainable fallback. The research operating point favours stable event timing; the demonstration operating point favours minimal reaction delay. The demo overlay in Fig. 5 shows the same signal chain in operator-oriented form: raw and stable stage labels, P(active_brazing), trigger state, HOLD

TEMPERATURE recommendation, inference time, and the ROI rectangle.



Fig. 5. Demonstration overlay for the active_brazing stage: ROI rectangle, raw and stable stage labels, $P(\text{active_brazing})$, neural trigger state and HOLD TEMPERATURE recommendation.

E. Latency and CPU-friendly variants

Table V collects full-pipeline latency. Standard ResNet18 at 224×224 misses the 50 ms target on CPU: total mean ≈ 117 ms, p95 ≈ 165 ms, with the network forward pass dominating. MobileNetV3 Small at the same input reaches total mean 25.04 ms and p95 38.92 ms, comfortably within the target. ResNet18 at 64×64 reaches 30.64 ms / 44.77 ms; after fine-tuning, p95 stabilises around 39.77 ms with usable quality. On GPU even the standard ResNet18 reaches p95 7.85 ms, with most of the time spent outside the network.

TABLE V. PER-FRAME LATENCY ON THE FULL PROCESSING PIPELINE

Model and mode	Device	Input	Total mean, ms	Total p95, ms	Model mean, ms	p95 < 50 ms
ResNet18	CPU	224×224	117.40	164.98	109.60	no
MobileNetV3 Small	CPU	224×224	25.04	38.92	16.95	yes
ResNet18	CPU	64×64	30.64	44.77	24.79	yes
ResNet18	GPU / CUDA	224×224	6.50	7.85	2.95	yes

All reported latency values correspond to single-frame, batch-size-1 measurements obtained in the experimental CPU/GPU environment used in this study. These timings are useful for comparing model variants under the same conditions. Before deployment, however, absolute latency should be measured again on the target industrial computer, since the final hardware and integration stack may change the real response time.

Table VI reports the quality-latency trade-off of CPU-friendly variants on the 10 FPS test set. Reducing the ResNet18 input from 128×128 to 64×64 monotonically lowers latency; after fine-tuning, ResNet18 64×64 reaches accuracy 0.9174 and macro-F1 0.8328 within p95 39.77 ms. MobileNetV3 Small after exploratory evolutionary tuning of its training hyper-parameters reaches comparable latency at somewhat lower per-frame quality but remains attractive because of its

smaller checkpoint and design for resource-limited environments.

TABLE VI. CPU-FRIENDLY VARIANTS ON THE 10 FPS TEST SET.

Model and input	Accuracy	Macro-F1	Weighted-F1	Active brazing F1	Total mean, ms	Total p95, ms
ResNet18, 128×128	0.8331	0.7559	0.8500	0.6158	43.76	65.26
ResNet18, 96×96	0.8580	0.7739	0.8715	0.6576	36.22	53.79
ResNet18, 64×64	0.8990	0.7967	0.9033	0.6534	26.16	39.77
ResNet18, 64×64 fine-tuned	0.9174	0.8328	0.9128	0.6410	26.16	39.77
MobileNet V3 Small, evo best, 224×224	0.8822	0.7820	0.8803	0.5714	21.69	34.75

Component-level analysis shows that the OpenCV trigger, neural trigger logic and state machine are negligible against the network forward pass; the dominant factors are architecture and input resolution. Two deployment regimes follow directly from these numbers: a qualitative regime with ResNet18 at 224×224 on GPU and a CPU-friendly regime that selects between ResNet18 64×64 (higher per-frame quality, slightly larger checkpoint) and MobileNetV3 Small (smaller checkpoint, comparable latency, marginally lower active_brazing recall). The numbers are reported for the experimental environment; deployment on a target industrial computer would require a repeat benchmark and may benefit from model compression and edge-inference optimisation.

V. DISCUSSION

The gap between the OpenCV baseline and ResNet18 confirms that hand-crafted features register visual change in the ROI but do not reliably separate flux_activation from early active_brazing. The remaining errors are concentrated at adjacent stage boundaries, where the visual signature changes continuously rather than discretely. The state machine addresses this operator-facing problem: it reduces full-test macro-F1 by about 0.02 relative to majority smoothing, but it produces a monotone and technologically valid stage sequence. Without MVI_6265, it remains close to smoothing on per-frame metrics and improves active_brazing F1 to 0.8333 with recall 0.7143.

MVI_6265 acts as a stress test: pronounced highlights and ambiguous flux dynamics imitate neighbouring stages and can provoke premature transitions. The gap between full-set and no-hard-case metrics, macro-F1 0.8290 against 0.9234 with smoothing, is therefore driven mainly by one difficult scenario. This points to adaptive ROI, image-quality control, additional viewpoints, and confidence-aware hard-case logic as the next improvements.

Latency results show a viable deployment path. The qualitative regime relies on GPU and ResNet18 at 224×224 . The CPU-friendly regime trades a small amount of per-frame quality for a 4–5 $\times$ reduction in forward-pass time and meets

the 50 ms target with MobileNetV3 Small or ResNet18 64×64 after fine-tuning. The most fragile metric is active_brazing recall on CPU-friendly models, where small visual cues are diluted by aggressive down-sampling; this is the value to watch when porting the prototype.

Limitations remain. The dataset is moderate and collected under similar lighting and camera conditions, so the absolute numbers do not extrapolate to arbitrary equipment. Annotation is purely visual and not synchronised with thermocouple or pyrometer readings; active_brazing_started here marks the visually observed start of active brazing rather than a strictly measured solidus or liquidus crossing — adequate for research on the visual channel, but it must be aligned with thermal telemetry before any closed-loop use. The state machine is non-returning by design: a feature in regular operation but a weakness if a transition is confirmed by mistake; confidence-based or multi-criterion confirmation is the natural next step. The latency benchmark is bound to the test environment and must be repeated on the target industrial hardware.

VI. CONCLUSION

A machine-vision module for induction brazing of spacecraft waveguide paths has been designed and evaluated. It decomposes the process into four visually distinguishable stages, classifies each ROI frame with a convolutional network, stabilises the operator-facing stage with a technologically constrained state machine, and generates an event signal active_brazing_started that drives a HOLD TEMPERATURE recommendation. The OpenCV baseline reaches macro-F1 0.5809 after smoothing. ResNet18 on 10 FPS reaches macro-F1 0.8209, or 0.9175 when MVI_6265 is excluded. State machine yields accuracy 0.9581 and macro-F1 0.9155 on regular recordings; the neural trigger detects the event on all test videos with mean absolute time error 0.70 s. MobileNetV3 Small and ResNet18 64×64 meet the 50 ms-per-frame latency target on CPU; the GPU regime keeps full-resolution ResNet18 well below 10 ms.

Future work will extend the recording set, synchronise video with temperature and inductor-power telemetry, add automatic ROI estimation, test model compression and edge-inference optimisation, and validate confidence-aware hard-case logic on an industrial bench.

REFERENCES

- [1] S. K. Zlobin, M. M. Mikhnev, V. D. Laptinok, Yu. N. Seregin, A. N. Bocharov, V. S. Tynchenko, Yu. P. Dubets, B. B. Dolgoplov, "Automated equipment and technology for brazing of spacecraft waveguide paths," *Vestnik of the Reshetnev Siberian State Aerospace University*, no. 4(56), pp. 219–229, 2014
- [2] N. F. Lashko, S. V. Lashko, *Brazing of Metals*, Moscow: Mashinostroyeniye, 1988
- [3] S. K. Zlobin, V. D. Laptinok, M. M. Mikhnev, R. V. Zaytsev, "Application of induction heating for brazing of waveguide-distribution path elements made of aluminium alloys," *Reshetnev Readings*, vol. 1, pp. 14–15, 2013.
- [4] A. V. Murugin, Yu. N. Seregin, V. S. Tynchenko, A. N. Bocharov, O. A. Bocharova, *Automated brazing by induction heating: monograph*, Krasnoyarsk: Reshetnev SibSU Publishing, 2022
- [5] V. S. Tynchenko, V. E. Petrenko, A. V. Milov, "Control of induction brazing on the basis of indirect temperature measurements," *Systems of Control and Information Technologies*, no. 4(78), pp. 50–54, 2019.
- [6] A. V. Milov, V. S. Tynchenko, A. V. Murugin, "Influence of flux on the accuracy of measurements during induction brazing of aluminium waveguide paths," *Modern Technologies. System Analysis. Modelling*, vol. 60, no. 4, pp. 38–46, 2018
- [7] Z. Li, Y. Yan, X. Wang, Y. Ge, L. Meng, "A survey of deep learning for industrial visual anomaly detection," *Artificial Intelligence Review*, vol. 58, Article 279, 2025.
- [8] V. S. Shukla, A. S. Shukla, S. P. S. K., S. S. Shukla, "A systematic survey: role of deep learning-based image anomaly detection in industrial inspection contexts," *Frontiers in Robotics and AI*, 2025, DOI: 10.3389/frobt.2025.1554196.
- [9] R. Szeliski, *Computer Vision: Algorithms and Applications*, 2nd ed., Cham: Springer, 2022.
- [10] I. Goodfellow, Y. Bengio, A. Courville, *Deep Learning*, Cambridge: MIT Press, 2016.
- [11] K. He, X. Zhang, S. Ren, J. Sun, "Deep Residual Learning for Image Recognition," *Proc. IEEE Conf. on Computer Vision and Pattern Recognition*, pp. 770–778, 2016.
- [12] A. Paszke et al., "PyTorch: An Imperative Style, High-Performance Deep Learning Library," *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [13] G. Bradski, "The OpenCV Library," *Dr. Dobb's Journal of Software Tools*, 2000.
- [14] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, Q. V. Le, H. Adam, "Searching for MobileNetV3," *Proc. IEEE/CVF Int. Conf. on Computer Vision*, pp. 1314–1324, 2019.
- [15] D. Ngo, H.-C. Park, B. Kang, "Edge Intelligence: A Review of Deep Neural Network Inference in Resource-Limited Environments," *Electronics*, vol. 14, no. 12, Article 2495, 2025
- [16] V. S. Tynchenko, S. O. Kurashkin, "Automated system of optimal control of induction brazing of spacecraft waveguide paths," *Instruments and Systems. Control, Inspection, Diagnostics*, no. 1, pp. 22–34, 2021
- [17] A. V. Milov, V. S. Tynchenko, S. O. Kurashkin, V. V. Tynchenko, V. V. Kukartsev, V. V. Bukhtoyarov, R. Sergienko, V. A. Kukartsev, K. A. Bashmur, "The use of collections of artificial neural networks to improve the control quality of the induction soldering process," *Sensors*, vol. 21, no. 12, p. 4199, 2021
- [18] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [19] I. Loshchilov, F. Hutter, "Decoupled Weight Decay Regularization," *Int. Conf. on Learning Representations*, 2019
- [20] C. Shorten, T. M. Khoshgoftaar, "A survey on Image Data Augmentation for Deep Learning," *Journal of Big Data*, vol. 6, no. 60, 2019